

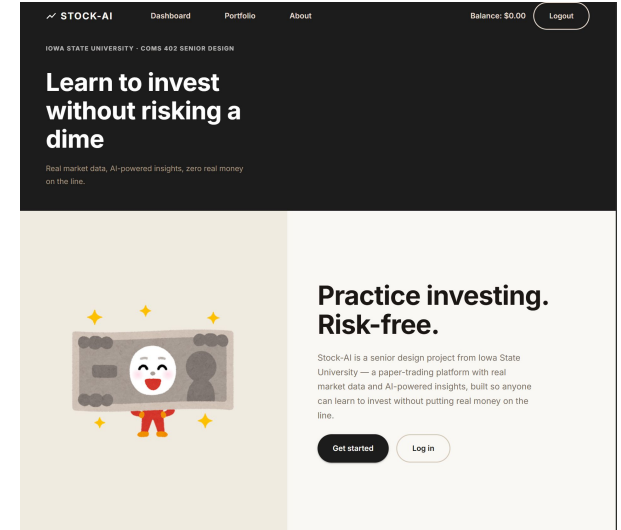
Stock-AI Demo #3

UG_HF_03

Andrew Butler, Matthew He, Ethan Roe, Aaron Trelstad

What is Stock-AI?

- Learning stock trading is difficult
- Trading & trading platforms are expensive
- Create Free learning platform
 - Mock trades, eliminating financial risk
 - AI assistance (ask questions, provide recommendations)



Technologies in use

Backend

- MariaDB
- FastAPI Python framework
- Podman for containerized deployment

Frontend

- React.js web framework
- Material UI component library

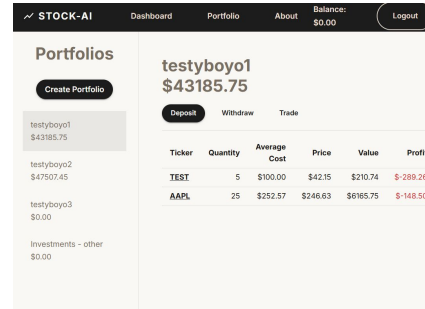
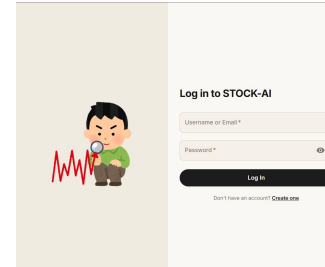
External Resources

- OpenAI API
- yfinance



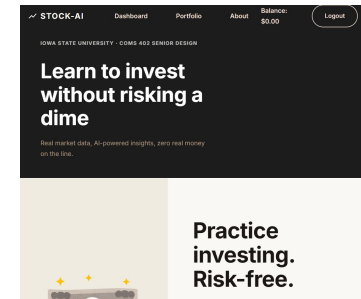
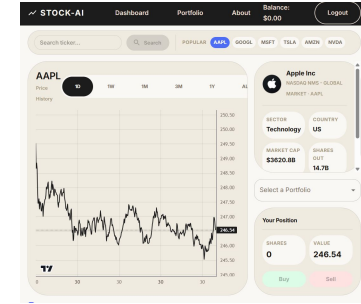
Stock-AI Features

- Secure login + authentication
- User Portfolios w/ value tracking
- Stock Dashboard
- AI Chat assistant
- AI Trading recommendations
- Competitive Leagues

The Portfolios page shows a list of portfolios on the left and a detailed view of 'testyboy01' on the right. The portfolio value is \$43185.75. Below the value are buttons for 'Deposit', 'Withdraw', and 'Trade'. A table lists the holdings.

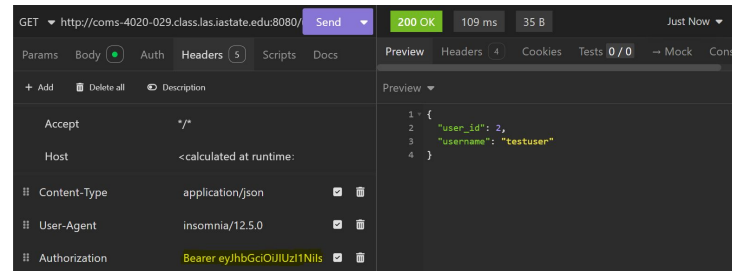
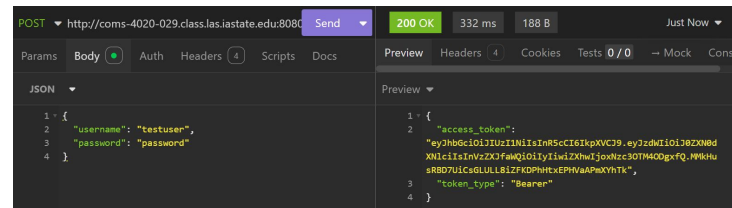
Portfolios						
Create Portfolio						
testyboy01	\$43185.75					
testyboy02	\$47507.45					
testyboy03	\$0.00					
Investments - other	\$0.00					

testyboy01		\$43185.75				
		Deposit	Withdraw	Trade		
Ticker	Quantity	Average Cost	Price	Value	Profit	
TEST	5	\$100.00	\$42.15	\$210.74	\$-289.26	
AAPL	25	\$252.57	\$246.63	\$6165.75	\$-148.50	



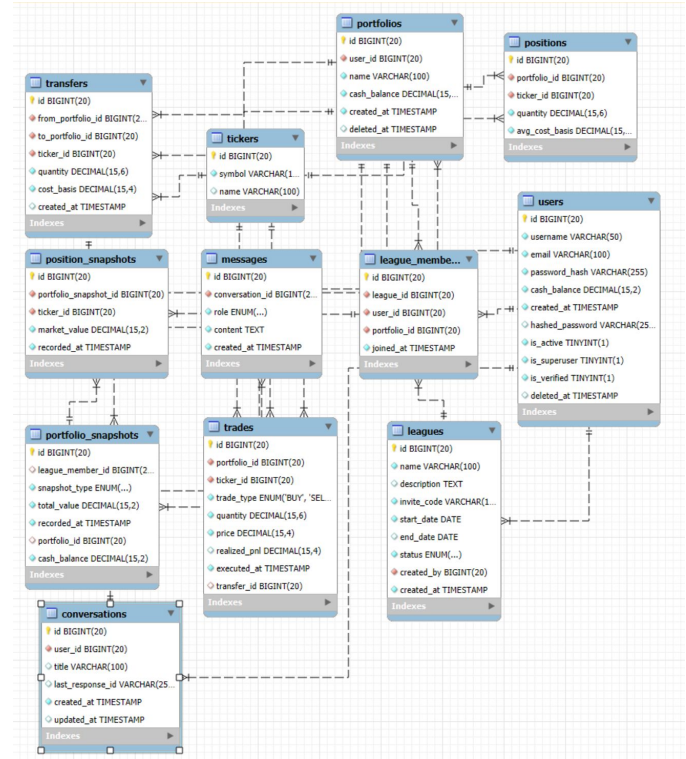
Security + Authentication

- Password hashing
- Login generates Json Web Token (JWT)
 - Gives token
 - Contains user ID, username
 - Expires after 60 minutes
 - Endpoints that are user specific require valid JWT
- Suboptimal: Implemented manually
 - Attempted refactor to 'FastAPI Users' library
 - Custom vs library User class conflict
 - Rebuild of database relations + endpoints
 - No added functionality



Database: stored information

- Lots of relations between stored data
- We store:
 - User information
 - Portfolio information
 - Portfolio snapshots (state at point in time)
 - Held positions
 - Specific information for Stock-AI features
- We do not store:
 - Historical stock pricing
 - Specific stock data (e.g., company description)
 - Why store when available with APIs?



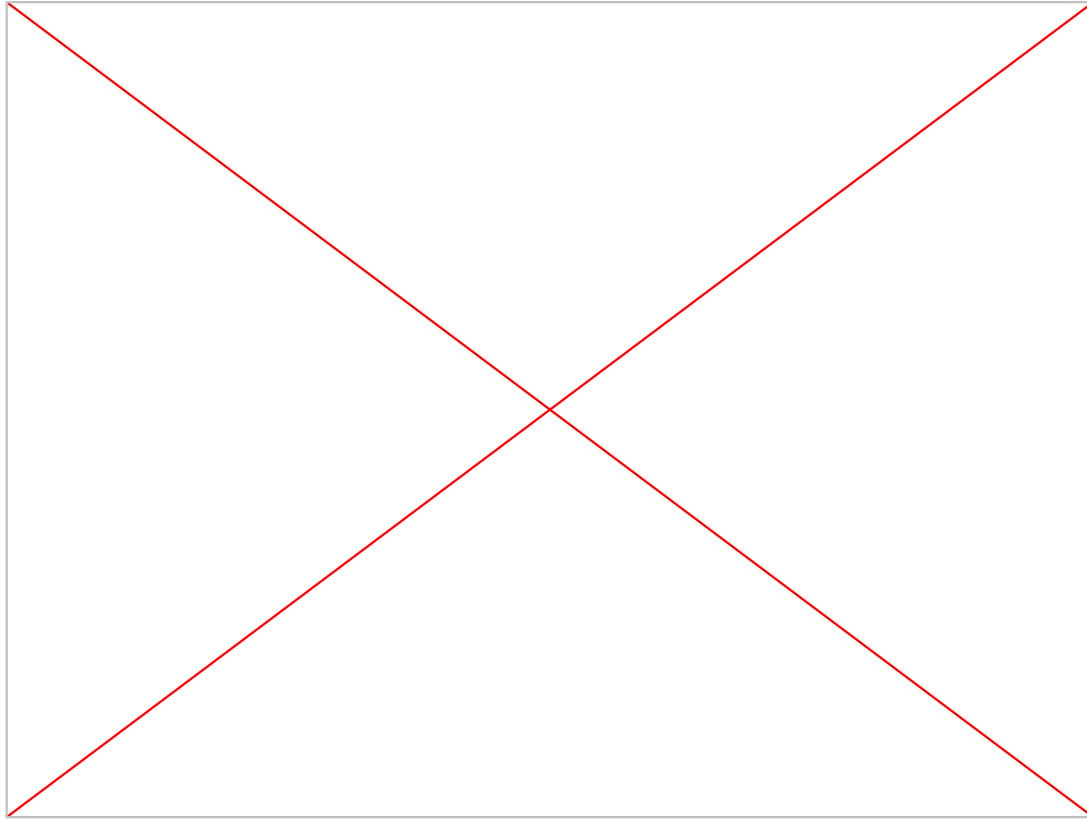
Portfolios, Users: Soft delete

Do not fully remove database entry on deletion

- Add field to mark as deleted
- Preserves related DB entries
- Makes recovery possible

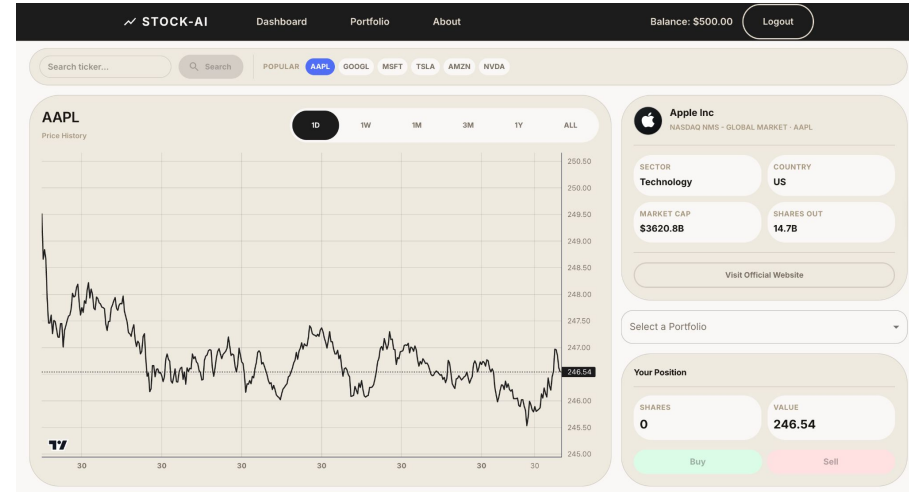
	id	user_id	name	cash_balance	created_at	deleted_at
▶	1	1	Investments Main	877.30	2026-02-24 19:08:36	NULL
	2	1	RenameTest	0.00	2026-02-24 19:08:41	2026-04-14 02:36:09

Users Demo



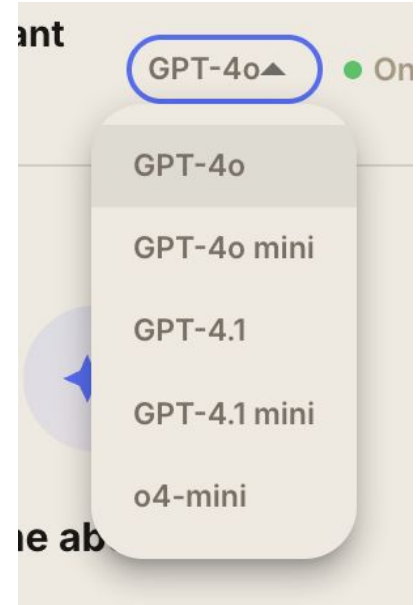
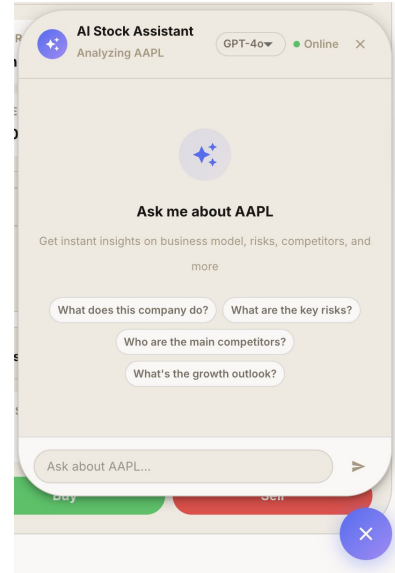
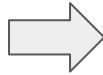
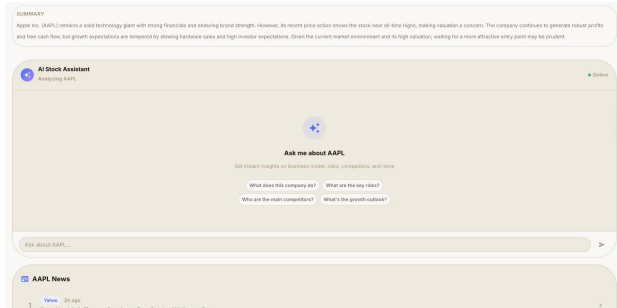
Stock Dashboard

- Search for any stock
- Visualize stock data
- Company information
- Market news



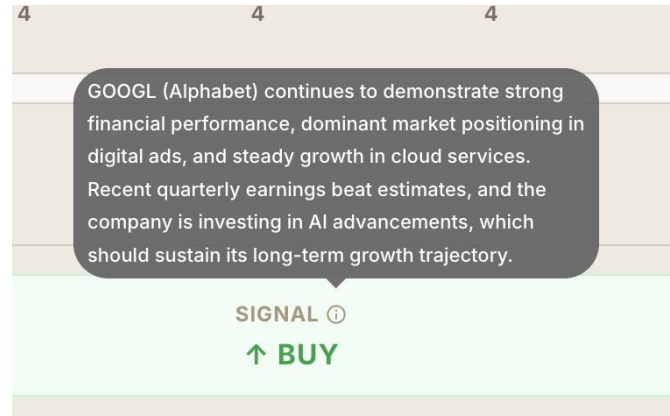
AI Chat Assistant

- Remembers conversation history
- Prompt engineering to keep on track
- Model selection (OpenAI)

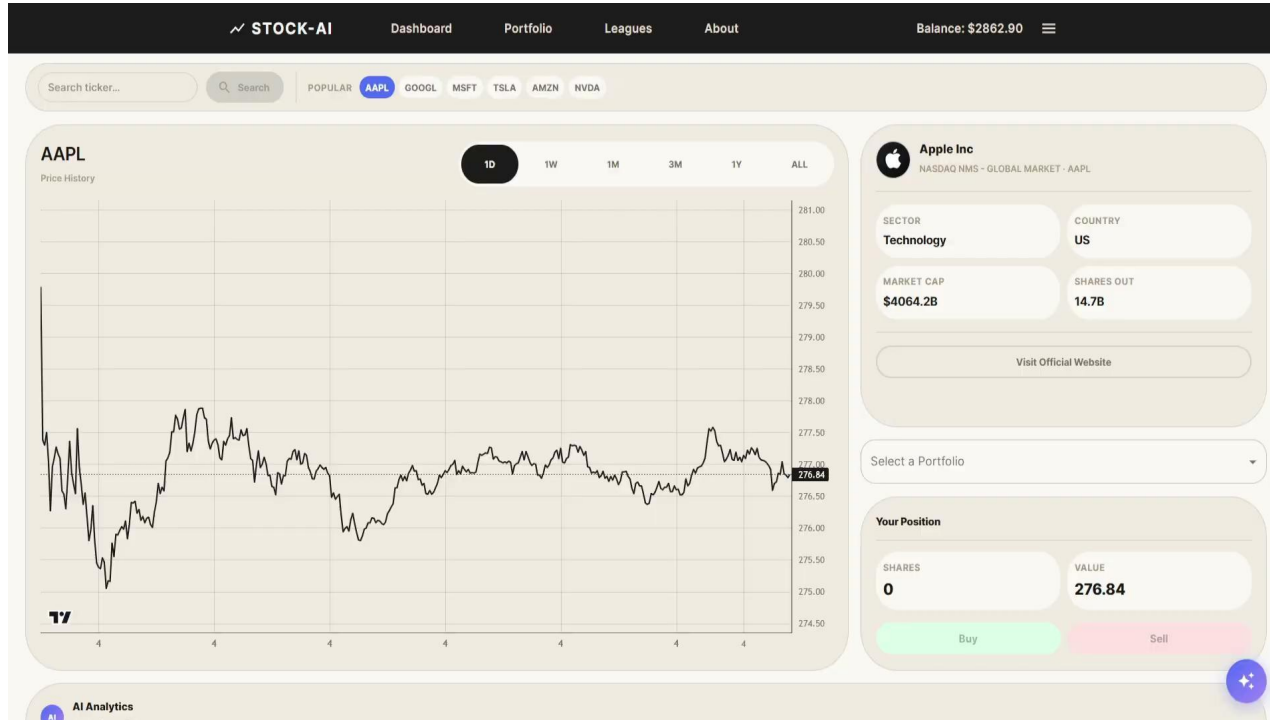


AI Trading Recommendations

- Buy/Sell signal
- Risk analysis
- Summaries
- Explanations



Dashboard Demo



User Portfolios

- User balance & Portfolio balance handled separately
- Organization of owned positions
- Trading done on portfolio basis
- Can transfer positions between portfolios
- Soft deletion

STOCK-AI Dashboard Portfolio Leagues About Balance: \$1362.90

Portfolios

Create Portfolio

Tech Investments \$5033.77

Misc Investments \$1500.00

Rename Delete

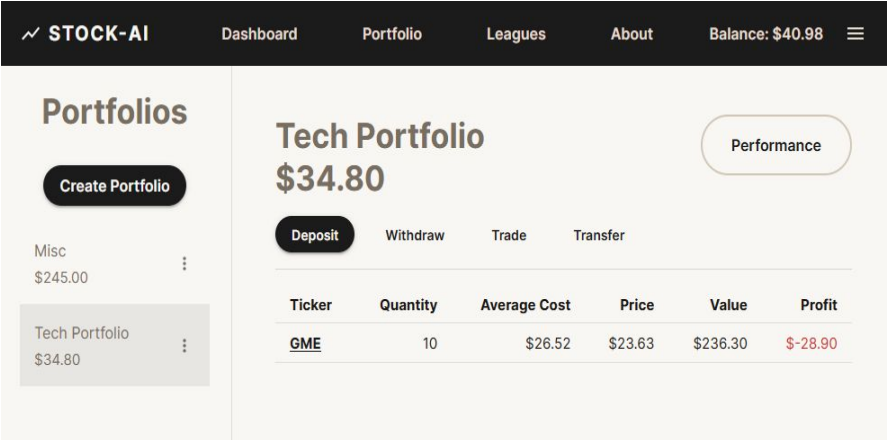
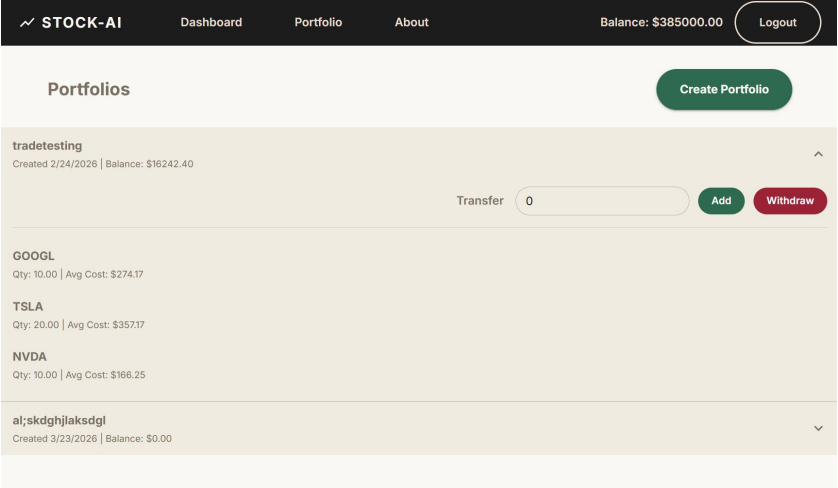
Tech Investments \$5033.77

Performance

Deposit Withdraw Trade Transfer

Ticker	Quantity	Average Cost	Price	Value	Profit
HPE	30	\$28.73	\$29.46	\$883.95	+\$22.11
MSFT	3	\$413.71	\$410.82	\$1232.46	-\$8.67

User Portfolios Design Improvements



User Portfolios Demo

STOCK-AI

DashboardPortfolioLeaguesAbout

Balance: \$4517.74

Portfolios

Create Portfolio

Tech Investments
\$4189.41

Tech Investments
\$4189.41

Tech Investments
\$4189.41

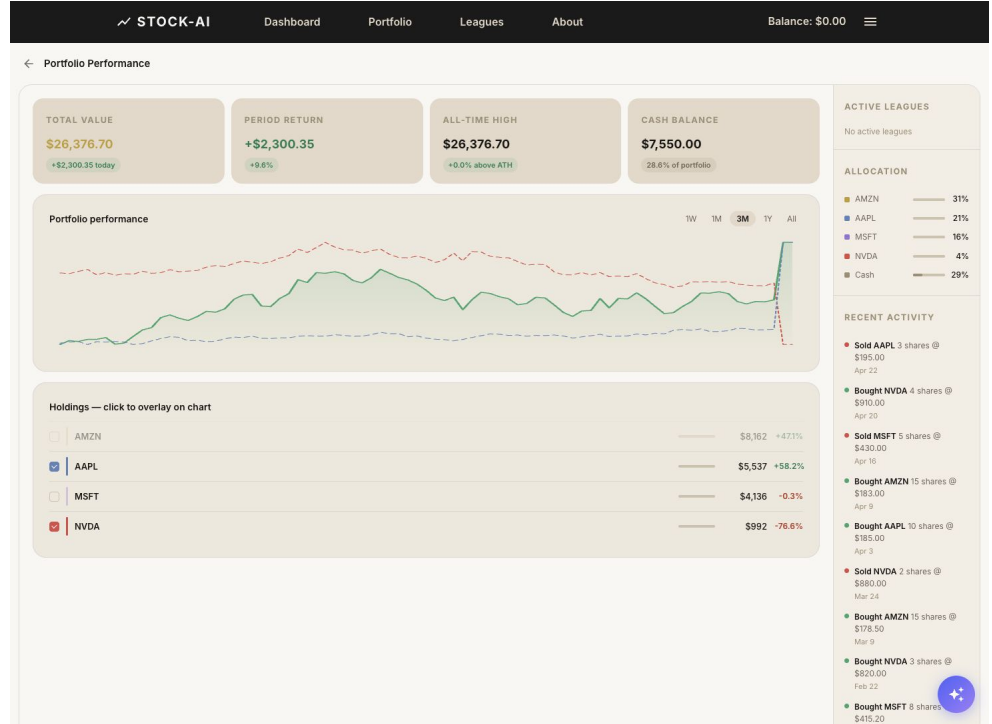
Performance

DepositWithdrawTradeTransfer

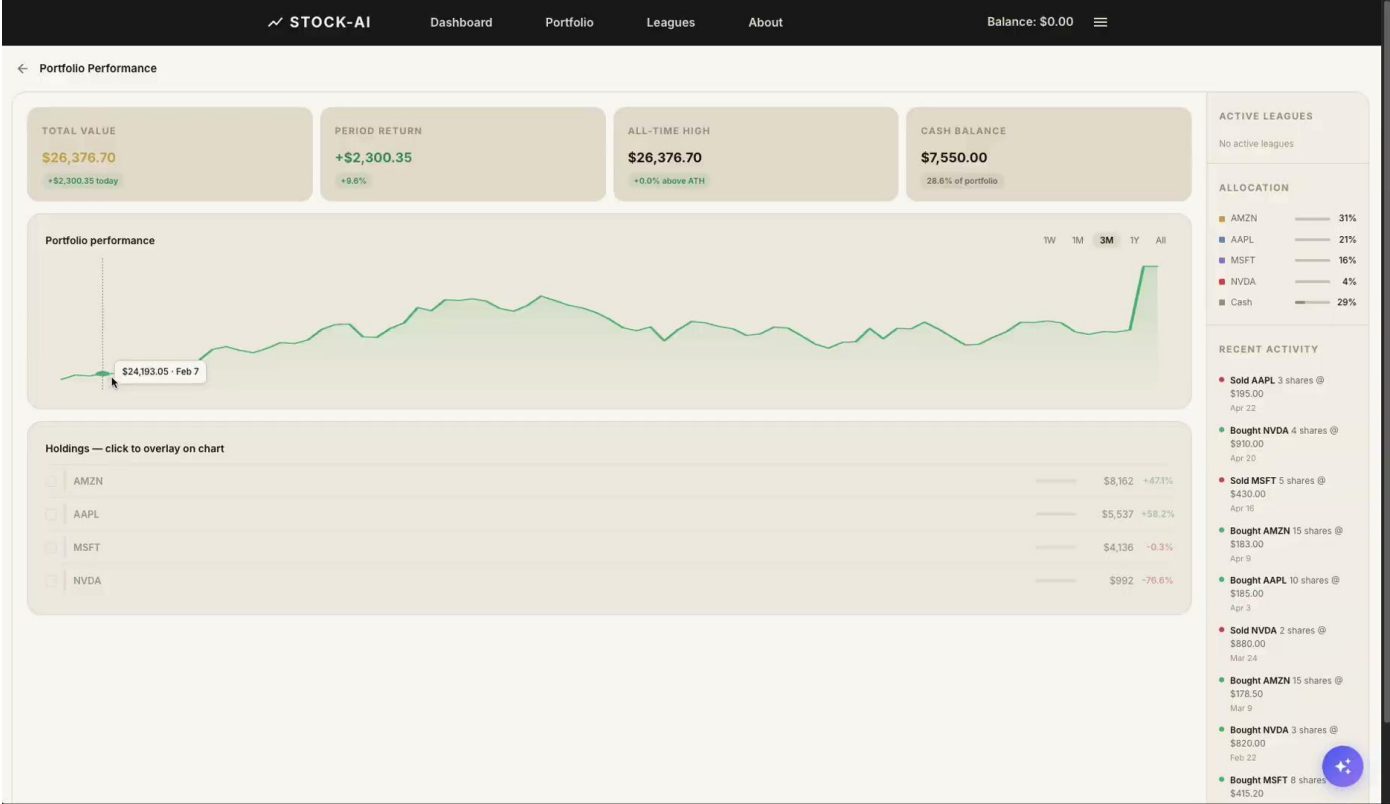
Ticker	Quantity	Average Cost	Price	Value	Profit
HPE	45	\$28.73	\$28.71	\$1291.95	\$-0.90

User Portfolios - Performance

- Value snapshot daily
- Trade history (bottom right)
- Stock allocation breakdown

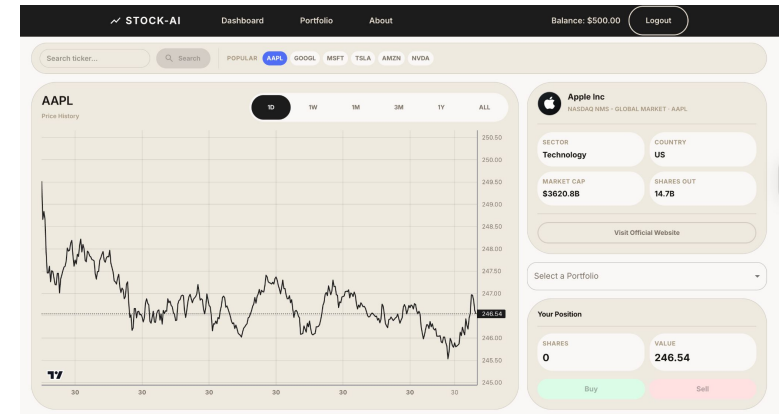
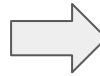
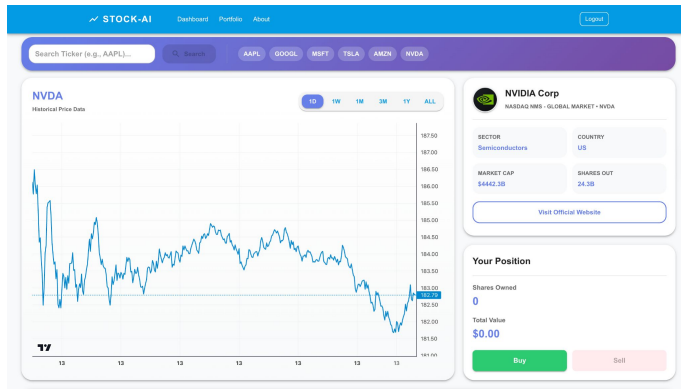
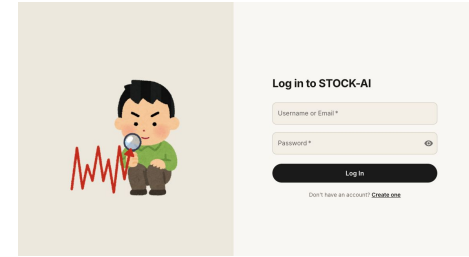


Portfolio Performance Demo



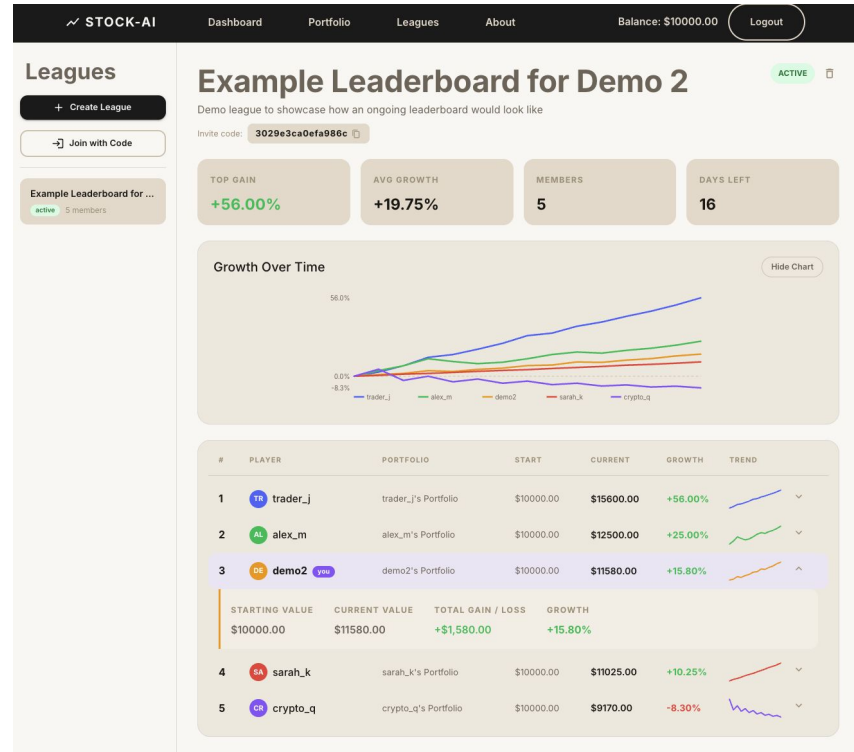
UI Design

- Originally bland, high contrast
- Moved to calm color palette
- Clipart



Competitive Leagues

- Completely optional
- Boost engagement



Competitive Leagues Demo - Expandable Rows + Deletion

STOCK-AI

DashboardPortfolioLeaguesAbout

Balance: \$7500.00

+ Create League

Join with Code

Example Leaderboard for ...

ended 5 members

Friends

ended 1 member

Demo 2

ended 3 members

testvideo

ended 2 members

Example Leaderboard for Demo 2

Demo league to showcase how an ongoing leaderboard would look like

Invite code: 3029e3c0defa986c

TOP GAIN

+56.00%

AVG GROWTH

+19.75%

MEMBERS

5

DAYS LEFT

0

Growth Over Time

Hide Chart

#	PLAYER	PORTFOLIO	START	CURRENT	GROWTH	TREND
1	trader_j	trader_j's Portfolio	\$10000.00	\$15600.00	+56.00%	
2	alex_m	alex_m's Portfolio	\$10000.00	\$12500.00	+25.00%	
3	demo2	demo2's Portfolio	\$10000.00	\$11580.00	+15.80%	
4	sarah_k	sarah_k's Portfolio	\$10000.00	\$11025.00	+10.25%	
5	crypto_q	crypto_q's Portfolio	\$10000.00	\$9170.00	-8.30%	

Leagues

+ Create League

Join with Code

demo league

pending 1 member

Example Leaderboard for ...

ended 5 members

Friends

ended 1 member

Demo 2

ended 5 members

testvideo

ended 2 members

Friends

This is a league for my friends

Invite code: b8c6fc414b95a908

TOP GAIN

+0.00%

AVG GROWTH

+0.00%

MEMBERS

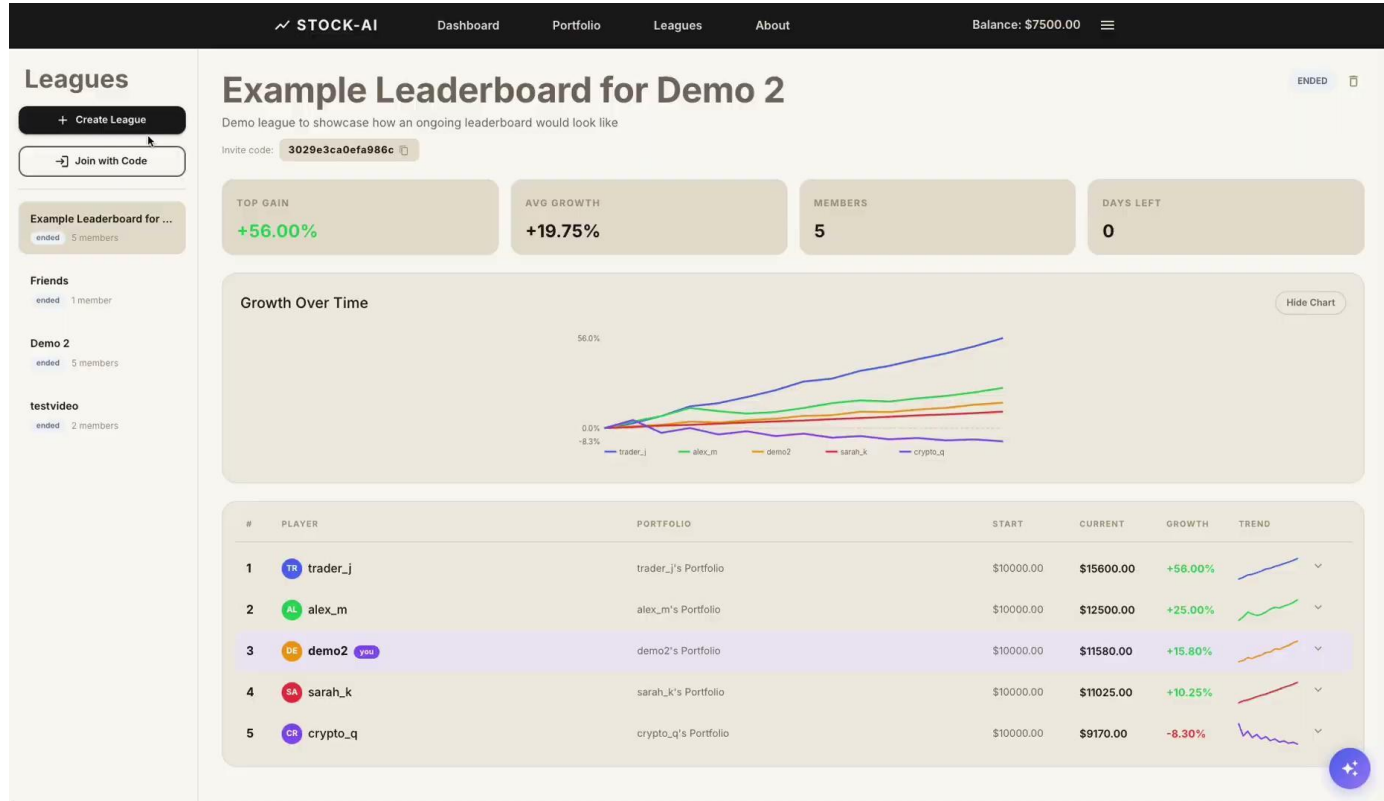
1

DAYS LEFT

0

#	PLAYER	PORTFOLIO	START	CURRENT	GROWTH	TREND
No entries yet.						

League Demo - Creation



League Demo - User Joining

STOCK-AI

DashboardPortfolioLeaguesAbout

Balance: \$0.00

Leagues

+ Create League

Join with Code

testvideo
ended 2 members

testvideo

testttttt

Invite code: a7be9aece469a998

TOP GAIN
+0.00%

AVG GROWTH
+0.00%

MEMBERS
2

DAYS LEFT
0

#

PLAYER

PORTFOLIO

START

CURRENT

GROWTH

TREND

No entries yet.

Development challenges

- Deployment + Connectivity
- UI Design
- AI Confusion
- Should have started with Users library

Summary of Contributions

- Andrew Butler

- Backend deployment
- Users + Auth backend
- Portfolios backend
- Stock trading backend

- Ethan Roe

- Users + Auth frontend
- Portfolios frontend
- Stock trading frontend

- Matthew He

- About page
- Leagues
- Portfolio performance
- Daily snapshots (cron job)

- Aaron Trelstad

- Dashboard
- AI Chatbot
- AI Analysis